

From Fisher Information to Adaptive Experimental Design: Statistical Optimality in Quantum State Tomography

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- ① Sequential estimation and adaptive experimental design
- ② Fisher information and optimal design
- ③ Quantum state tomography as a statistical problem
- ④ Adaptive measurement design for quantum tomography
- ⑤ Optimality theory and simulation results

A simple example: estimating a Gaussian mean

Assume

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2),$$

where σ_0^2 is known.

The estimator is

$$\hat{\mu}_n = \bar{X}_n.$$

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We have

$$\sqrt{n}(\bar{X}_n - \mu) \sim N(0, \sigma_0^2).$$

The variance of the estimator is controlled by the information contained in the data.

From Gaussian models to general parametric models

For a regular parametric model

$$X \sim f_\theta, \quad \theta \in \mathbb{R}^p,$$

the MLE satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, \mathcal{I}(\theta)^{-1}).$$

The Fisher information matrix is

$$\mathcal{I}(\theta) = \mathbb{E}_\theta[\nabla_\theta \log f_\theta(X) \nabla_\theta \log f_\theta(X)^T].$$

MLE under sequential experimental design

Now suppose the experiment can be chosen adaptively.

At each step, $X_t|a_t \sim f_{\theta,a_t}$, where $a_t \in \{1, \dots, k\}$ is the selected experiment.

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If $\bar{\pi}_n(a) \rightarrow \pi_a, a = 1, \dots, k$, then the accumulated information satisfies

$$\frac{1}{n} J_n(\theta) \rightarrow \mathcal{I}^\pi(\theta) := \sum_{a=1}^k \pi_a \mathcal{I}_a(\theta), \text{ where } J_n(\theta) = \sum_{t=1}^n \mathcal{I}_{a_t}(\theta).$$

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Asymptotic behavior of the MLE

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The central question:

How should we collect data to maximize the information about θ ?

Why adaptive experimental design?

In many experiments, we can choose the next experiment.

Examples:

- medical experiments: choose the next treatment;
- quantum experiments: choose the next measurement.

The choice should depend on previous observations.

$$(a_1, X_1), \dots, (a_n, X_n) \longrightarrow a_{n+1}.$$

Sequential experimental design

At time t :

- 1 Choose action

$$a_t = \phi_t(H_{t-1}),$$

where

$$H_{t-1} = \{(a_i, X_i) : i < t\}.$$

- 2 Observe

$$X_t | a_t \sim f_{\theta, a_t}.$$

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Goal:

Design a sequential policy whose estimator is optimal.

Static design and averaged Fisher information

Suppose actions are selected with proportions

$$\pi = (\pi_1, \dots, \pi_k) \in \Delta_k.$$

The averaged Fisher information is

$$\mathcal{I}^\pi(\theta) = \sum_{a=1}^k \pi_a \mathcal{I}_a(\theta).$$

Different choices of π lead to different estimation accuracy.

Why optimal design is nontrivial

For multidimensional parameters,

$$\text{Cov}(\hat{\theta}) \approx \frac{1}{n} \mathcal{I}^{\pi}(\theta)^{-1}.$$

Matrices are partially ordered:

$$A \succeq B$$

does not imply every pair is comparable.

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Therefore we need scalar criteria preserving the Loewner ordering.

The classical A-optimal criterion minimizes average variance:

$$\pi_A^* = \arg \min_{\pi \in \Delta_k} \text{tr}\{\mathcal{I}^\pi(\theta)^{-1}\}.$$

Interpretation:

- smaller total estimation variance;
- optimal average precision.

Classical optimal design, see Pukelsheim (2006).

Loss induced optimal design

Consider a loss satisfying

$$d_D^2(\theta, \theta + \Delta\theta) = \Delta\theta^T G_D(\theta) \Delta\theta + o(\|\Delta\theta\|^2).$$

Combining with

$$\sqrt{n}(\hat{\theta}_n - \theta) \approx N(0, \mathcal{I}^{\bar{\pi}_n}(\theta)^{-1}),$$

gives

$$\mathbb{E} d_D^2(\hat{\theta}_n, \theta) \approx \frac{1}{n} \operatorname{tr} (G_D(\theta) [\mathcal{I}^{\bar{\pi}_n}(\theta)]^{-1}).$$

See Li and Zhao (2025) for the sequential optimality theory.

Loss optimal design criterion

The optimal allocation minimizes the asymptotic estimation risk

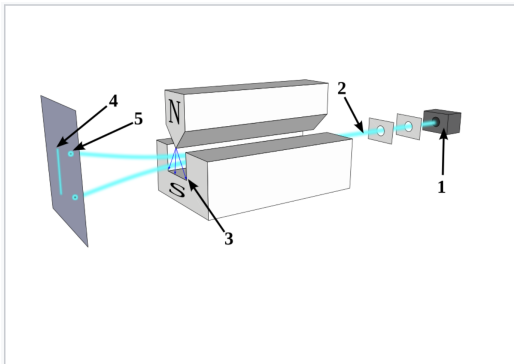
$$\Phi_D(\pi; \theta) = \text{tr} \left(G_D(\theta) [\mathcal{I}^\pi(\theta)]^{-1} \right).$$

The criterion balances two components:

- $\mathcal{I}^\pi(\theta)$: information obtained from the experimental design;
- $G_D(\theta)$: local geometry of the loss function.

The optimal design depends on the statistical objective and the uncertainty structure of the parameter.

Stern–Gerlach experiment: measuring quantum spin



Stern–Gerlach experiment: Silver atoms travelling through an inhomogeneous magnetic field, and being deflected up or down depending on their spin; (1) furnace, (2) beam of silver atoms, (3) inhomogeneous magnetic field, (4) classically expected result, (5) observed result

A magnetic field gradient separates spin-1/2 particles into two outcomes:

spin up (+1) and spin down (-1)

along the measurement direction.

From spin measurement to Born's rule

A spin-1/2 state is represented by a Bloch vector

$$\rho = \frac{1}{2}(I + \theta^\top \sigma), \quad \theta \in \mathbb{R}^3, \quad \|\theta\| \leq 1.$$

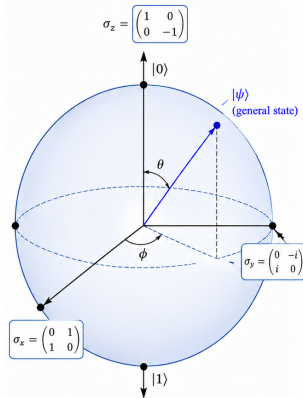
A Stern–Gerlach apparatus with magnetic-field gradient direction $u = (u_x, u_y, u_z)$, $\|u\| = 1$ measures the spin observable

$$\sigma_u = u_x \sigma_X + u_y \sigma_Y + u_z \sigma_Z.$$

The Born rule gives

$$P(+1 \mid u, \rho) = \text{tr} \left(\rho \frac{I + \sigma_u}{2} \right) = \frac{1 + \theta^\top u}{2}.$$

The angle between the Bloch vector and the magnetic-field gradient determines the measurement probability.



Pure states (points on the sphere)

Any qubit pure state $\leftrightarrow$
a unit vector on the Bloch sphere

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

$$(0 \leq \theta \leq \pi, 0 \leq \phi < 2\pi)$$

Density matrices (inside the sphere)

Any qubit density matrix can be written as

$$\rho = \frac{1}{2} (I + \theta^\top \sigma)$$

where $\sigma = (\sigma_x, \sigma_y, \sigma_z)^\top$ and
 $\theta = (x, y, z)^\top \in \mathbb{R}^3$ with $\|\theta\| \leq 1$.

- $\|\theta\| = 1$: pure states (surface)
- $\|\theta\| < 1$: mixed states (inside)

Quantum state estimation

The unknown object is the density matrix

$$\rho \in \mathbb{C}^{2 \times 2}.$$

Constraints:

$$\rho \succeq 0, \quad \text{tr}(\rho) = 1.$$

The task:

$$\mathbb{E} d_D^2(\hat{\rho}, \rho).$$

What does tomography mean?

Tomography reconstructs an object from projections.

Examples:

- medical CT reconstructs a body from X-ray projections;
- quantum tomography reconstructs ρ from measurement probabilities.

Measurements as statistical experiments

A quantum measurement setting a is described by a POVM

$$\{Q_{a,b} : b \in [r_a]\},$$

where

$$Q_{a,b} \succeq 0, \quad \sum_{b=1}^{r_a} Q_{a,b} = I.$$

Here

$$a \in [k], \quad r_a \geq 2.$$

For a state $\rho(\theta)$, Born's rule gives

$$p_{a,b}(\theta) = P(X = b|a) = \text{tr}(\rho(\theta)Q_{a,b}).$$

A quantum experiment produces classical random outcomes.

Quantum statistical model

We use a local parametrization of the density matrix

$$\rho(\theta), \quad \theta \in \Theta \subset \mathbb{R}^d.$$

A measurement setting a produces classical outcomes with probabilities

$$p_{a,b}(\theta) = \text{tr}(\rho(\theta)Q_{a,b}).$$

Therefore, quantum tomography becomes a classical statistical estimation problem:

$$\text{measurement data} \longrightarrow \hat{\theta}_n \longrightarrow \hat{\rho}_n.$$

A single measurement provides only partial information about ρ , so selecting informative measurements is essential.

Local geometry of quantum state loss

We measure estimation error through a distance between states:

$$d_D(\rho(\theta + \Delta\theta), \rho(\theta)), \quad D \in \{F, B\}.$$

For full-rank states $\rho(\theta) \succ 0$, both Frobenius and Bures distances admit the local expansion

$$d_D^2(\theta, \theta + \Delta\theta) = \Delta\theta^T G_D(\theta) \Delta\theta + o(\|\Delta\theta\|^2).$$

- Frobenius loss:

$$d_F(\rho, \sigma) = \|\rho - \sigma\|_F, \quad G_F = \frac{1}{2} I_d.$$

- Bures loss:

$$d_B(\rho, \sigma)^2 = 2 - 2 \operatorname{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}},$$

where $G_B(\theta)$ is the positive definite Bures metric matrix.

Likelihood and maximum likelihood estimation

Given observations (a_t, b_t) , define counts

$$N_{a,b} = \sum_{t=1}^n 1\{a_t = a, b_t = b\}.$$

The likelihood is

$$L(\rho) = \prod_{a,b} \{\text{tr}(\rho Q_{a,b})\}^{N_{a,b}}.$$

The constrained MLE is

$$\hat{\rho}_n = \arg \max_{\rho \succeq 0, \text{tr } \rho = 1} L(\rho).$$

The design problem is to choose future measurements to improve this estimator.

Fisher information for quantum measurements

Use a local parametrization

$$\rho(\theta), \quad \theta \in \mathbb{R}^d.$$

Define

$$s_i(a, b) = \text{tr}(\sigma_i Q_{a,b}),$$

and

$$p_{a,b}(\theta) = \text{tr}(\rho(\theta) Q_{a,b}).$$

For one measurement setting a ,

$$[\mathcal{I}_a(\theta)]_{ij} = \frac{1}{4} \sum_b \frac{s_i(a, b) s_j(a, b)}{p_{a,b}(\theta)}.$$

Loss-optimal experimental design

For a static design

$$\pi \in \Delta_k,$$

the averaged Fisher information is

$$\mathcal{I}^\pi(\theta) = \sum_{a=1}^k \pi_a \mathcal{I}_a(\theta).$$

The optimal allocation solves

$$\pi_D^*(\theta) = \arg \min_{\pi \in \Delta_k} \text{tr} \left(G_D(\theta) \{ \mathcal{I}^\pi(\theta) \}^{-1} \right).$$

The objective depends on both:

- information provided by measurements;
- geometry of the loss.

Adaptive measurement selection

For adaptive experiments,

$$J_n = \sum_{t=1}^n \mathcal{I}_{a_t}(\hat{\theta}_n)$$

is the accumulated Fisher information.

If

$$\bar{\pi}_n(a) = \frac{1}{n} \sum_{t=1}^n 1\{a_t = a\},$$

then

$$\frac{1}{n} J_n \approx \mathcal{I}^{\bar{\pi}_n}(\theta).$$

For an efficient estimator,

$$\mathbb{E} d_D^2(\hat{\theta}_n, \theta) \approx \frac{1}{n} \text{tr} (G_D(\theta) [\mathcal{I}^{\bar{\pi}_n}(\theta)]^{-1}).$$

Greedy information design

The goal is to approximate the optimal allocation π_D^* sequentially.

$$a_{n+1} = \arg \min_a \operatorname{tr} \left[G_D(\hat{\theta}_n)(J_n + \mathcal{I}_a)^{-1} \right].$$

This is the GI0 rule.

Using

$$(J + \mathcal{I}_a)^{-1} \approx J^{-1} - J^{-1}\mathcal{I}_aJ^{-1},$$

we obtain GI1:

$$a_{n+1} = \arg \max_a \operatorname{tr}(G_D(\hat{\theta}_n)J_n^{-1}\mathcal{I}_aJ_n^{-1}).$$

Convergence to the loss-optimal measurement allocation

Theorem: asymptotic optimality of adaptive design

Fix a loss function $D \in \{A, B\}$, and run the corresponding D -GI0 or D -GI1 adaptive measurement rule.

Let $\bar{\pi}_n$ be the empirical measurement frequency after n single-shot measurements. Define the optimal static design

$$\pi_D^*(\theta) = \arg \min_{\pi \in \Delta_k} \Phi_D(\pi; \theta),$$

where

$$\Phi_D(\pi; \theta) = \text{tr} \left(G_D(\theta) \{ \mathcal{I}^\pi(\theta) \}^{-1} \right).$$

If the minimizer is unique, then

$$\bar{\pi}_n \longrightarrow \pi_D^*(\theta) \quad \mathbb{P}_\theta\text{-a.s.}$$

Local asymptotic minimax optimality

Theorem: local asymptotic minimax lower bound

Consider any regular adaptive estimation procedure T_n .

For the local alternatives

$$\theta_n = \theta + \frac{h}{\sqrt{n}},$$

the asymptotic risk satisfies

$$\begin{aligned} \sup_{|F| < \infty} \liminf_{n \rightarrow \infty} \sup_{h \in F} \mathbb{E}_{\theta + h/\sqrt{n}} \left[nd_D^2 \left(\rho(T_n), \rho \left(\theta + \frac{h}{\sqrt{n}} \right) \right) \right] \\ \geq \inf_{\pi \in \Delta_k} \text{tr} (G_D(\theta) [\mathcal{I}^\pi(\theta)]^{-1}). \end{aligned}$$

- The right-hand side is the optimal risk achievable by any measurement allocation.
- No regular adaptive procedure can outperform this bound.

Optimality of adaptive GI design

Theorem: asymptotic efficiency of adaptive MLE

Let $\hat{\theta}_n$ be the MLE obtained under the loss-optimal GI0 or GI1 measurement strategy. The empirical measurement frequencies converge to the optimal allocation: $\bar{\pi}_n \rightarrow \pi_D^*(\theta)$, where

$$\pi_D^*(\theta) = \arg \min_{\pi \in \Delta_k} \text{tr} (G_D(\theta)[\mathcal{I}^\pi(\theta)]^{-1}).$$

Consequently,

$$\lim_{n \rightarrow \infty} n \mathbb{E}_\theta d_D^2 \left(\rho(\hat{\theta}_n), \rho(\theta) \right) = \inf_{\pi \in \Delta_k} \text{tr} (G_D(\theta)[\mathcal{I}^\pi(\theta)]^{-1}).$$

Adaptive design learns the optimal experiment allocation and achieves the local minimax lower bound.

Oracle design versus adaptive design

The optimal allocation depends on the unknown state θ .

The oracle design assumes θ is known:

$$\pi_D^*(\theta) = \arg \min_{\pi \in \Delta_k} \text{tr} (G_D(\theta)[\mathcal{I}^\pi(\theta)]^{-1}).$$

In practice, θ is unknown.

The adaptive GI procedure estimates θ sequentially and learns the oracle allocation from data.

adaptive design $\longrightarrow$ oracle performance.

Simulation: one-qubit Pauli library

We compare measurement strategies:

- uniform allocation;
- G10 adaptive design;
- G11 adaptive design;
- oracle optimal design.

For the one-qubit Pauli library, the adaptive procedures approach the oracle performance.

Simulation: One-qubit measurement libraries

For a unit measurement direction $u \in \mathbb{R}^3$,

$$\sigma_u = u_x \sigma_x + u_y \sigma_y + u_z \sigma_z, \quad \|u\|_2 = 1.$$

The corresponding projective measurement has POVM elements $Q_{u,\pm} = \frac{I \pm \sigma_u}{2}$, corresponding to spin outcomes $+1$ and -1 .

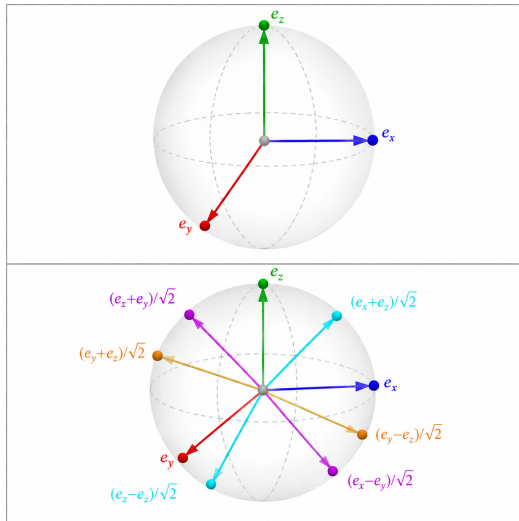
We compare two one-qubit measurement libraries:

- **Pauli library**

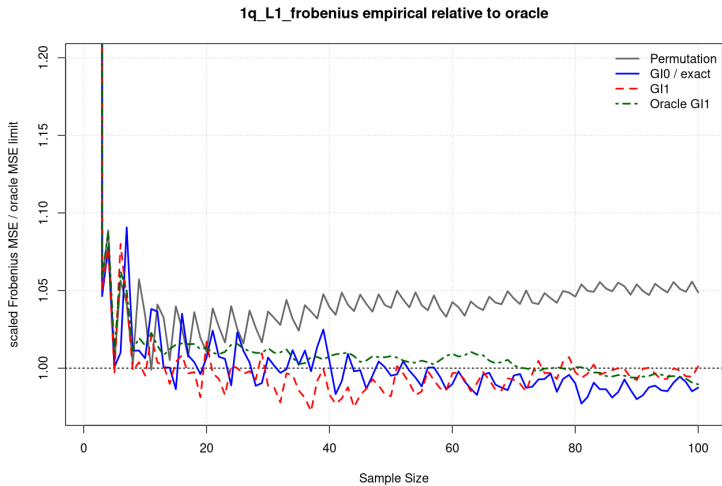
$$u \in \{e_x, e_y, e_z\}.$$

- **Nine-setting projective library**

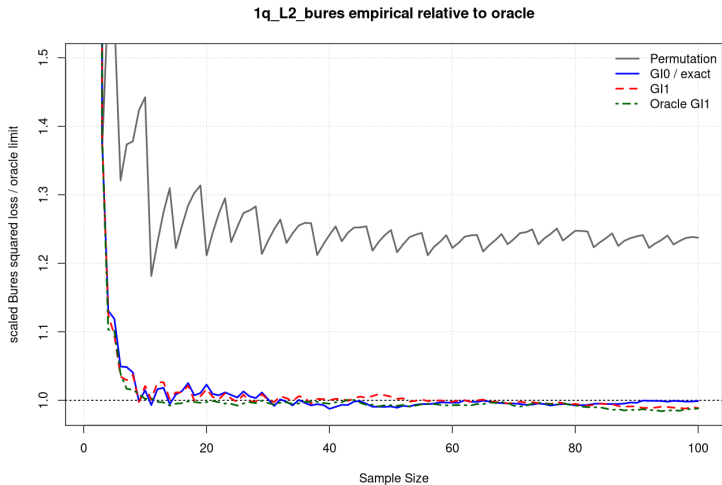
$$u \in \left\{ e_x, e_y, e_z, \frac{e_x \pm e_y}{\sqrt{2}}, \frac{e_x \pm e_z}{\sqrt{2}}, \frac{e_y \pm e_z}{\sqrt{2}} \right\}.$$



Simulation results



Simulation results: nine-setting measurements



- Quantum tomography is a statistical estimation problem.
- Measurements are experimental choices controlled by the researcher.
- Fisher information connects measurements and estimation accuracy.
- Loss geometry determines the optimal design criterion.
- Adaptive GI methods achieve asymptotic optimality.

References



F. Pukelsheim.

Optimal Design of Experiments.

SIAM, 2006.



X. Li and H. Zhao.

Globally-Optimal Greedy Active Sequential Estimation.

IEEE Transactions on Information Theory, 71(5):3871–3924, 2025.